

GATE DA (Data Science & AI) Practice Question Bank - Set 6

7 Topics · Previous-Year-Style + New Practice Questions · Full Solutions

This set covers seven topics: Standard Distributions I (Binomial, Poisson, Uniform), Classification Metrics (confusion matrix, Precision/Recall/F1, ROC-AUC), SQL Basics (SELECT/WHERE/ORDER BY/basic JOIN), Time, Speed & Distance, Logarithmic Loops Complexity, Binary Search (3 problems + complexity variants), and Uninformed Search -- DLS & IDS.

SAME HONESTY POLICY AS BEFORE: GATE DA has run only in 2024, 2025 and 2026, and it is a non-coding, MCQ/MSQ/NAT written exam. Every "old" question below is tagged with its real provenance:

[VERIFIED PYQ] -- a genuine, well-documented past GATE question (year and paper stated). Where the source is GATE CSE rather than GATE DA specifically, that is stated explicitly.

[GATE-style] -- not a verbatim past question (either no verified PYQ exists on that exact sub-point, or the original figure/data could not be reliably reproduced), but written to match real GATE DA difficulty, phrasing, and mark-weightage.

Section B ("New Practice Questions") in every topic is freshly written GATE-level practice -- none of these are claimed to be past questions.

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Topic 1: Standard Distributions I – Binomial, Poisson, Uniform

Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE DA 2024, Question 7]

Q1. [NAT · 1 mark]

A family has three children, where each child independently has probability $1/2$ of being a boy and $1/2$ of being a girl. Using the binomial PMF $P(X=k) = C(n,k)p^k(1-p)^{n-k}$, find the probability that the family has exactly 2 girls.

Answer: 0.375

Solution: $n=3$, $p=1/2$, $k=2$. $P(2 \text{ girls}) = C(3,2)(1/2)^2(1/2)^1 = 3 \times (1/4) \times (1/2) = 3/8 = 0.375$.

[VERIFIED PYQ — GATE DA 2024, Question 11]

Q2. [MCQ · 1 mark]

Consider the following statements: (i) The mean and variance of a Poisson random variable are equal. (ii) For a standard normal random variable, the mean is zero and the variance is one. Which ONE of the following options is correct?

- (A) Both (i) and (ii) are true
- (B) (i) is true and (ii) is false
- (C) (ii) is true and (i) is false
- (D) Both (i) and (ii) are false

Answer: (A)

Solution: For Poisson(λ), both the mean and variance equal λ -- statement (i) is true. For the standard normal distribution $N(0,1)$, mean=0 and variance=1 by definition -- statement (ii) is also true. So both statements are true.

[GATE-style]

Q3. [NAT · 1 mark]

$X \sim \text{Binomial}(n=6, p=0.5)$. Find $P(X=3)$.

Answer: 0.3125

Solution: $P(X=3) = C(6,3)(0.5)^6 = 20 \times 0.015625 = 0.3125$.

[GATE-style]

Q4. [NAT · 2 marks]

$X \sim \text{Poisson}(\lambda=4)$. Find $P(X=0)$ (correct to 4 decimals).

Answer: 0.0183

Solution: $P(X=0) = e^{-4} \approx 0.0183$.

[GATE-style]

Q5. [MCQ · 1 mark]

For $X \sim \text{Uniform}(a,b)$, the PDF $f(x)$ is:

- (A) $1/(b-a)$ for $a \leq x \leq b$
- (B) $(b-a)$ for $a \leq x \leq b$
- (C) $1/(b+a)$
- (D) $x/(b-a)$

Answer: (A)

[GATE-style]

Q6. [NAT · 1 mark]

$X \sim \text{Uniform}(0,10)$. Find $E[X]$.

Answer: 5

Solution: $E[X] = (a+b)/2 = (0+10)/2 = 5$.

[GATE-style]

Q7. [MCQ · 1 mark]

The Poisson distribution is often used to approximate the Binomial distribution when:

- (A) n is small and p is close to 0.5
- (B) n is large and p is small, with $np=\lambda$ held moderate
- (C) n equals p
- (D) The distribution is continuous

Answer: (B)

[GATE-style]

Q8. [MSQ · 2 marks]

Which of the following are TRUE about the Binomial(n,p) distribution?

- (A) It models the number of successes in n independent Bernoulli(p) trials
- (B) Its mean is np
- (C) Its variance is $np(1-p)$
- (D) It requires p to change from trial to trial

Answer: (A), (B), (C)

Solution: (D) is false -- the standard binomial model requires p to remain CONSTANT across all trials.

[GATE-style]

Q9. [NAT · 2 marks]

$X \sim \text{Poisson}(\lambda=2)$. Find $P(X=2)$ (correct to 4 decimals).

Answer: 0.2707

Solution: $P(X=2) = e^{-2}(2)^2/2! = 0.1353 \times 4/2 = 0.1353 \times 2 = 0.2707$.

[GATE-style]

Q10. [MCQ · 1 mark]

For a continuous Uniform(a,b) distribution, the variance is:

- (A) $(b-a)/2$
- (B) $(b-a)^2/12$
- (C) $(b+a)^2/12$
- (D) $(b-a)^2/4$

Answer: (B)

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 2 marks]

$X \sim \text{Binomial}(n=5, p=0.4)$. Find $P(X=2)$ (correct to 4 decimals).

Answer: 0.3456

Solution: $P(X=2) = C(5,2)(0.4)^2(0.6)^3 = 10 \times 0.16 \times 0.216 = 0.3456$.

[New Practice]

Q12. [NAT · 2 marks]

$X \sim \text{Poisson}(\lambda=3)$. Find $P(X=1)$ (correct to 4 decimals).

Answer: 0.1494

Solution: $P(X=1) = e^{-3}(3) \approx 0.0498 \times 3 = 0.1494$.

[New Practice]

Q13. [MCQ · 1 mark]

$X \sim \text{Uniform}(2,10)$. Find $P(4 < X < 8)$.

- (A) 0.5
- (B) 0.4
- (C) 0.6
- (D) 0.25

Answer: (A)

Solution: $P = (8-4)/(10-2) = 4/8 = 0.5$.

[New Practice]

Q14. [NAT · 1 mark]

$X \sim \text{Binomial}(n=8, p=0.5)$. Find $E[X]$.

Answer: 4

Solution: $E[X] = np = 8 \times 0.5 = 4$.

[New Practice]

Q15. [NAT · 1 mark]

Using the same $X \sim \text{Binomial}(n=8, p=0.5)$, find $\text{Var}(X)$.

Answer: 2

Solution: $\text{Var}(X) = np(1-p) = 8 \times 0.5 \times 0.5 = 2$.

[New Practice]

Q16. [MSQ · 2 marks]

Which of the following are TRUE about the Poisson distribution?

- (A) It models the number of rare events occurring in a fixed interval of time or space
- (B) Its mean equals its variance
- (C) It requires a fixed, known number of trials n , like the binomial
- (D) The parameter λ represents the average rate of occurrence

Answer: (A), (B), (D)

Solution: (C) is false -- unlike the binomial, the Poisson distribution does not require a fixed number of discrete trials.

[New Practice]

Q17. [NAT · 1 mark]

$X \sim \text{Uniform}(5,15)$. Find $\text{Var}(X)$ (correct to 2 decimals).

Answer: 8.33

Solution: $\text{Var}(X) = (15-5)^2/12 = 100/12 \approx 8.33$.

[New Practice]

Q18. [MCQ · 1 mark]

If $X \sim \text{Binomial}(n,p)$ and n is very large while p is very small, such that $np=\lambda$ remains constant, X can be well-approximated by:

- (A) $\text{Uniform}(0,\lambda)$
- (B) $\text{Normal}(0,1)$
- (C) $\text{Poisson}(\lambda)$
- (D) $\text{Exponential}(\lambda)$

Answer: (C)

[New Practice]

Q19. [NAT · 2 marks]

$X \sim \text{Poisson}(\lambda=5)$. Find $P(X=0)$ (correct to 4 decimals).

Answer: 0.0067

Solution: $P(X=0) = e^{-5} \approx 0.0067$.

[New Practice]

Q20. [MCQ · 1 mark]

For a fair six-sided die roll, modelled as a discrete $\text{Uniform}\{1,\dots,6\}$ distribution, the variance is:

- (A) 2.92
- (B) 3.5
- (C) 2.5
- (D) 3.0

Answer: (A)

Solution: $\text{Var} = (n^2-1)/12 = (36-1)/12 = 35/12 \approx 2.92$.

Topic 2: Classification Metrics - Confusion Matrix, P/R/F1, ROC-AUC

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [MCQ · 1 mark]

In a binary classification confusion matrix, a False Positive (FP) refers to:

- (A) The model correctly predicts the positive class
- (B) The model incorrectly predicts the positive class when the actual class is negative
- (C) The model correctly predicts the negative class
- (D) The model incorrectly predicts the negative class when the actual class is positive

Answer: (B)

[GATE-style]

Q2. [MCQ · 1 mark]

Precision is defined as:

- (A) $TP/(TP+FN)$
- (B) $TP/(TP+FP)$
- (C) $TN/(TN+FP)$
- (D) $(TP+TN)/(TP+TN+FP+FN)$

Answer: (B)

[GATE-style]

Q3. [MCQ · 1 mark]

Recall (sensitivity) is defined as:

- (A) $TP/(TP+FN)$
- (B) $TP/(TP+FP)$
- (C) $TN/(TN+FN)$
- (D) $(TP+TN)/(TP+TN+FP+FN)$

Answer: (A)

[GATE-style]

Q4. [NAT · 1 mark]

A binary classifier produces the confusion matrix: $TP=40$, $FP=10$, $FN=20$, $TN=30$. Find the Precision.

Answer: 0.8

Solution: Precision = $40/(40+10) = 40/50 = 0.8$.

[GATE-style]

Q5. [NAT · 1 mark]

Using the same confusion matrix ($TP=40$, $FP=10$, $FN=20$, $TN=30$), find the Recall (correct to 4 decimals).

Answer: 0.6667

Solution: Recall = $40/(40+20) = 40/60 \approx 0.6667$.

[GATE-style]

Q6. [NAT · 2 marks]

Using the same confusion matrix, find the F1 score (correct to 4 decimals).

Answer: 0.7273

Solution: $F1 = 2 \times P \times R / (P + R) = 2 \times 0.8 \times 0.6667 / (0.8 + 0.6667) = 1.0667 / 1.4667 \approx 0.7273$ (exactly $8/11$).

[GATE-style]

Q7. [NAT · 1 mark]

Using the same confusion matrix, find the Accuracy.

Answer: 0.7

Solution: $\text{Accuracy} = (TP + TN) / \text{Total} = (40 + 30) / 100 = 0.7$.

[GATE-style]

Q8. [MCQ · 1 mark]

The ROC curve plots:

- (A) Precision vs Recall
- (B) True Positive Rate (TPR) vs False Positive Rate (FPR) at various classification thresholds
- (C) Accuracy vs threshold
- (D) F1 score vs threshold

Answer: (B)

[GATE-style]

Q9. [MCQ · 1 mark]

An AUC (Area Under the ROC Curve) value of 0.5 indicates:

- (A) A perfect classifier
- (B) A classifier performing no better than random guessing
- (C) A classifier that is always wrong
- (D) An undefined result

Answer: (B)

[GATE-style]

Q10. [MSQ · 2 marks]

Which of the following are TRUE about classification metrics?

- (A) High precision means few false positives, relative to predicted positives
- (B) High recall means few false negatives, relative to actual positives
- (C) F1 score is the harmonic mean of precision and recall
- (D) Accuracy is always the best metric to use, regardless of class balance

Answer: (A), (B), (C)

Solution: (D) is false -- accuracy can be misleading on imbalanced datasets (e.g. a model predicting only the majority class can have high accuracy but be useless).

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

A binary classifier produces: TP=50, FP=5, FN=15, TN=30. Find the Precision (correct to 4 decimals).

Answer: 0.9091

Solution: Precision = $50/55 \approx 0.9091$.

[New Practice]

Q12. [NAT · 1 mark]

Using the same confusion matrix, find the Recall (correct to 4 decimals).

Answer: 0.7692

Solution: Recall = $50/65 \approx 0.7692$.

[New Practice]

Q13. [NAT · 2 marks]

Using the same confusion matrix, find the F1 score (correct to 4 decimals).

Answer: 0.8333

Solution: $F1 = \frac{2 \times (10/11) \times (10/13)}{[(10/11) + (10/13)]} = \frac{(200/143)}{(240/143)} = \frac{200}{240} = \frac{5}{6} \approx 0.8333$.

[New Practice]

Q14. [NAT · 1 mark]

Using the same confusion matrix, find the Accuracy.

Answer: 0.8

Solution: Accuracy = $(50+30)/100 = 0.8$.

[New Practice]

Q15. [MCQ · 1 mark]

The False Positive Rate (FPR), used in ROC curves, is defined as:

- (A) $FP/(FP+TN)$
- (B) $FP/(FP+TP)$
- (C) $TP/(TP+FN)$
- (D) $TN/(TN+FP)$

Answer: (A)

[New Practice]

Q16. [MCQ · 1 mark]

Which metric is most appropriate when False Negatives are particularly costly (e.g. missing a disease diagnosis)?

- (A) Precision
- (B) Recall
- (C) Specificity
- (D) AUC alone, ignoring all else

Answer: (B)

[New Practice]

Q17. [MSQ · 2 marks]

Which of the following are TRUE about the ROC-AUC metric?

- (A) AUC ranges from 0 to 1
- (B) A higher AUC generally indicates better discriminative ability between classes
- (C) AUC is threshold-independent, since it summarizes performance across all thresholds
- (D) AUC is only defined for multi-class classification, never binary

Answer: (A), (B), (C)

Solution: (D) is false -- ROC-AUC is a standard metric for BINARY classification (multi-class variants exist too, but the classic AUC-ROC is fundamentally a binary-classification metric).

[New Practice]

Q18. [NAT · 1 mark]

A confusion matrix has TP=25, FP=15, FN=5, TN=55. Find the Specificity, defined as $TN/(TN+FP)$ (correct to 4 decimals).

Answer: 0.7857

Solution: Specificity = $55/(55+15) = 55/70 \approx 0.7857$.

[New Practice]

Q19. [MCQ · 1 mark]

If a classifier predicts every single instance as positive, the Recall will be:

- (A) 0
- (B) 1 (100%), since no actual positives are missed (FN=0)
- (C) Undefined
- (D) Equal to precision, always

Answer: (B)

Solution: Predicting everything as positive means FN=0, so Recall = $TP/(TP+0) = 1$.

[New Practice]

Q20. [NAT · 2 marks]

Using the confusion matrix TP=25, FP=15, FN=5, TN=55, find the F1 score (correct to 4 decimals).

Answer: 0.7143

Solution: Precision = $25/40 = 0.625$. Recall = $25/30 \approx 0.8333$. $F1 = 2 \times 0.625 \times 0.8333 / (0.625 + 0.8333)$
 $= (25/24) / (35/24) = 25/35 = 5/7 \approx 0.7143$.

Topic 3: SQL Basics - SELECT / WHERE / ORDER BY / Basic JOIN

Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE CS 2015, Set 1]

Q1. [MCQ · 1 mark]

The SELECT operation in SQL is equivalent to which relational algebra operation?

- (A) The selection operation in relational algebra
- (B) The selection operation in relational algebra, except that SQL's SELECT retains duplicate rows
- (C) The projection operation in relational algebra
- (D) The projection operation in relational algebra, except that SQL's SELECT retains duplicate rows

Answer: (D)

Solution: SQL's SELECT clause (choosing columns) corresponds to the relational algebra PROJECTION operation (π), not selection (σ). However, unlike relational algebra's projection (which removes duplicate tuples under set semantics), SQL's SELECT retains duplicate rows by default (bag/multiset semantics) unless DISTINCT is explicitly used.

[VERIFIED PYQ — GATE CSE 2019, concept]

Q2. [MCQ · 2 marks]

Given schemas employees(emp-id, first-name, last-name, hire-date, dept-id, salary) and departments(dept-id, dept-name, manager-id, location-id), to display the last names and hire dates of the MOST RECENTLY HIRED employee in each department at location-id 1700, which SQL technique is essential?

- (A) A simple WHERE clause only, with no subquery
- (B) A subquery using MAX(hire-date) combined with GROUP BY dept-id, matched via IN against (dept-id, hire-date) pairs
- (C) ORDER BY hire-date DESC alone, with no grouping
- (D) DISTINCT alone, with no subquery

Answer: (B)

Solution: Since the maximum hire-date is needed PER department, a grouped subquery (SELECT dept-id, MAX(hire-date) ... GROUP BY dept-id) is required, and the outer query matches against these (dept-id, hire-date) pairs -- the essential technique in this classic GATE query pattern.

[GATE-style]

Q3. [MCQ · 1 mark]

The basic syntax order of an SQL SELECT statement's clauses is:

- (A) SELECT, FROM, WHERE, ORDER BY
- (B) FROM, SELECT, ORDER BY, WHERE
- (C) WHERE, SELECT, FROM, ORDER BY
- (D) ORDER BY, SELECT, FROM, WHERE

Answer: (A)

[GATE-style]

Q4. [MCQ · 1 mark]

Which SQL clause is used to filter individual rows based on a condition, BEFORE any grouping is applied?

- (A) HAVING
- (B) WHERE
- (C) ORDER BY
- (D) GROUP BY

Answer: (B)

[GATE-style]

Q5. [MCQ · 1 mark]

Which SQL clause is used to sort the result set of a query?

- (A) SORT BY
- (B) ORDER BY
- (C) GROUP BY
- (D) ARRANGE BY

Answer: (B)

[GATE-style]

Q6. [MCQ · 1 mark]

By default (with no ASC/DESC keyword specified), ORDER BY sorts results in:

- (A) Ascending order
- (B) Descending order
- (C) Random order
- (D) No particular order

Answer: (A)

[GATE-style]

Q7. [MCQ · 1 mark]

An INNER JOIN between tables A and B returns:

- (A) All rows from A, and matching rows from B, with NULLs where there is no match
- (B) Only the rows where there is a match in both A and B, based on the join condition
- (C) All rows from both A and B, regardless of match
- (D) Only rows from A that have no match in B

Answer: (B)

[GATE-style]

Q8. [MCQ · 1 mark]

A LEFT OUTER JOIN between tables A (left) and B (right) returns:

- (A) Only matching rows from both tables
- (B) All rows from A, plus matched rows from B (with NULLs in B's columns for unmatched A rows)
- (C) All rows from B, plus matched rows from A
- (D) Only rows from B with no match in A

Answer: (B)

[GATE-style]

Q9. [MSQ · 2 marks]

Which of the following are TRUE about basic SQL query clauses?

- (A) The WHERE clause cannot use aggregate functions like COUNT() or SUM() directly
- (B) The HAVING clause filters groups after GROUP BY has been applied, and CAN use aggregate functions
- (C) SELECT DISTINCT removes duplicate rows from the result
- (D) ORDER BY can only sort by columns that appear in the SELECT list

Answer: (A), (B), (C)

Solution: (D) is false -- most SQL implementations allow ORDER BY to reference columns not included in the SELECT list, as long as they belong to a table in the FROM clause.

[GATE-style]

Q10. [MCQ · 1 mark]

The SQL wildcard character used in a LIKE clause to match zero or more characters is:

- (A) ?
- (B) #
- (C) %
- (D) *

Answer: (C)

Section B — New Practice Questions (10)

[New Practice]

Q11. [MCQ · 1 mark]

Which query correctly retrieves all columns from table Employees where Salary is greater than 50000?

- (A) SELECT * FROM Employees WHERE Salary > 50000;
- (B) SELECT * FROM Employees HAVING Salary > 50000;
- (C) SELECT Employees WHERE Salary > 50000;
- (D) GET * FROM Employees IF Salary > 50000;

Answer: (A)

[New Practice]

Q12. [MCQ · 1 mark]

Which query sorts employees by Salary in descending order?

- (A) SELECT * FROM Employees ORDER BY Salary ASC;
- (B) SELECT * FROM Employees ORDER BY Salary DESC;
- (C) SELECT * FROM Employees SORT BY Salary DESC;
- (D) SELECT * FROM Employees GROUP BY Salary DESC;

Answer: (B)

[New Practice]

Q13. [MCQ · 2 marks]

Given tables Students(StudentID, Name) and Marks(StudentID, Score), which query correctly performs an inner join to list each student's name with their score?

- (A) SELECT S.Name, M.Score FROM Students S, Marks M;
- (B) SELECT S.Name, M.Score FROM Students S INNER JOIN Marks M ON S.StudentID = M.StudentID;
- (C) SELECT S.Name, M.Score FROM Students S UNION Marks M;
- (D) SELECT Name, Score FROM Students, Marks WHERE Name = Score;

Answer: (B)

[New Practice]

Q14. [NAT · 1 mark]

Table Products has 200 rows. The query "SELECT * FROM Products WHERE Price > 100" returns 45 rows. How many rows have Price ≤ 100?

Answer: 155

Solution: 200 – 45 = 155.

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following are valid uses of the WHERE clause?

- (A) Filtering rows based on a numeric comparison, e.g. WHERE age > 18
- (B) Filtering rows based on a string pattern using LIKE
- (C) Combining multiple conditions using AND / OR
- (D) Filtering aggregated group results, e.g. WHERE COUNT(*) > 5

Answer: (A), (B), (C)

Solution: (D) is false -- filtering based on an aggregate function's result must be done with HAVING, not WHERE.

[New Practice]

Q16. [MCQ · 1 mark]

What does the query "SELECT DISTINCT Department FROM Employees;" return?

- (A) All rows from Employees
- (B) A list of unique department values, with duplicates removed
- (C) The count of departments
- (D) An error, since DISTINCT cannot be used with SELECT

Answer: (B)

[New Practice]

Q17. [MCQ · 1 mark]

Which of the following correctly represents a basic multi-condition WHERE clause?

- (A) WHERE Age > 18 AND City = 'Delhi'
- (B) WHERE Age > 18, City = 'Delhi'
- (C) WHERE Age > 18; City = 'Delhi'
- (D) WHERE (Age > 18)(City = 'Delhi')

Answer: (A)

[New Practice]

Q18. [MCQ · 1 mark]

Which of the following correctly limits query results to only the top 5 rows after sorting, in common SQL dialects using LIMIT?

- (A) SELECT * FROM Employees ORDER BY Salary DESC LIMIT 5;
- (B) SELECT TOP FROM Employees LIMIT 5;
- (C) SELECT * FROM Employees WHERE ROWNUM STOP 5;
- (D) SELECT * FROM Employees MAX 5;

Answer: (A)

[New Practice]

Q19. [MCQ · 1 mark]

A RIGHT OUTER JOIN between tables A (left) and B (right) returns:

- (A) Only matching rows
- (B) All rows from B, plus matched rows from A (with NULLs in A's columns for unmatched B rows)
- (C) All rows from A, plus matched rows from B
- (D) Only rows from A with no match in B

Answer: (B)

[New Practice]

Q20. [NAT · 2 marks]

Table Orders has 500 rows, and table Customers has 100 rows. If every order references exactly one existing customer (a valid foreign key), how many rows will "SELECT * FROM Orders INNER JOIN Customers ON Orders.CustomerID = Customers.CustomerID;" return?

Answer: 500

Solution: Since the foreign key guarantees a match for every order, the inner join preserves all 500 order rows (each matching exactly one customer).

Topic 4: Time, Speed & Distance

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [NAT · 1 mark]

A car travels 300 km in 5 hours. Find its speed in km/h.

Answer: 60

Solution: Speed = Distance/Time = $300/5 = 60$ km/h.

[GATE-style]

Q2. [NAT · 1 mark]

A train travels at 80 km/h for 3.5 hours. Find the distance covered.

Answer: 280

Solution: Distance = Speed×Time = $80 \times 3.5 = 280$ km.

[GATE-style]

Q3. [MCQ · 2 marks]

A car covers a distance in 6 hours at 50 km/h. If it travels at 60 km/h instead, how long will it take to cover the same distance?

- (A) 5 hours
- (B) 4 hours
- (C) 5.5 hours
- (D) 6 hours

Answer: (A)

Solution: Distance = $50 \times 6 = 300$ km. New time = $300/60 = 5$ hours.

[GATE-style]

Q4. [NAT · 1 mark]

Convert a speed of 72 km/h into m/s.

Answer: 20

Solution: $72 \times (5/18) = 20$ m/s.

[GATE-style]

Q5. [NAT · 1 mark]

Convert a speed of 15 m/s into km/h.

Answer: 54

Solution: $15 \times (18/5) = 54$ km/h.

[GATE-style]

Q6. [MCQ · 2 marks]

Two trains start from the same point and travel in OPPOSITE directions at 60 km/h and 40 km/h respectively. After how many hours will they be 300 km apart?

- (A) 3
- (B) 2

- (C) 4
- (D) 5

Answer: (A)

Solution: Relative speed (opposite directions) = $60+40 = 100$ km/h. Time = $300/100 = 3$ hours.

[GATE-style]

Q7. [NAT · 2 marks]

A boat travels downstream at 20 km/h and upstream at 12 km/h. Find the speed of the boat in still water.

Answer: 16

Solution: Boat speed = $(\text{downstream} + \text{upstream})/2 = (20+12)/2 = 16$ km/h.

[GATE-style]

Q8. [NAT · 1 mark]

Using the previous question's data (downstream 20 km/h, upstream 12 km/h), find the speed of the current.

Answer: 4

Solution: Current speed = $(\text{downstream} - \text{upstream})/2 = (20-12)/2 = 4$ km/h.

[GATE-style]

Q9. [MSQ · 2 marks]

Which of the following are TRUE about time-speed-distance relationships?

- (A) Distance = Speed $\times$ Time
- (B) When two objects move toward each other, their relative speed is the SUM of their individual speeds
- (C) When two objects move in the same direction, their relative speed is the DIFFERENCE of their individual speeds
- (D) Speed is always measured in km/h only, never in other units

Answer: (A), (B), (C)

Solution: (D) is false -- speed can be measured in m/s, mph, knots, and other units too.

[GATE-style]

Q10. [MCQ · 2 marks]

A person walking at 5 km/h reaches a destination 12 minutes late. Walking at 6 km/h, they reach 8 minutes early. Find the distance to the destination.

- (A) 10 km
- (B) 12 km
- (C) 8 km
- (D) 15 km

Answer: (A)

Solution: Let t = scheduled time in hours. At 5 km/h: distance = $5(t+12/60)$. At 6 km/h: distance = $6(t-8/60)$. Setting equal: $5t+1 = 6t-0.8 \Rightarrow t=1.8$. Distance = $5(1.8+0.2) = 5 \times 2 = 10$ km.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

A cyclist covers 150 km in 6 hours. Find the speed in km/h.

Answer: 25

Solution: $150/6 = 25$ km/h.

[New Practice]

Q12. [NAT · 1 mark]

A car travels at 90 km/h for 2.5 hours. Find the distance covered.

Answer: 225

Solution: $90 \times 2.5 = 225$ km.

[New Practice]

Q13. [MCQ · 1 mark]

A journey takes 4 hours at 40 km/h. How long would it take at 80 km/h?

- (A) 2 hours
- (B) 3 hours
- (C) 2.5 hours
- (D) 1.5 hours

Answer: (A)

Solution: Distance = 160 km. Time at 80 km/h = $160/80 = 2$ hours.

[New Practice]

Q14. [NAT · 1 mark]

Convert a speed of 108 km/h into m/s.

Answer: 30

Solution: $108 \times (5/18) = 30$ m/s.

[New Practice]

Q15. [NAT · 1 mark]

Convert a speed of 25 m/s into km/h.

Answer: 90

Solution: $25 \times (18/5) = 90$ km/h.

[New Practice]

Q16. [MCQ · 2 marks]

Two cars start from the same point and travel in the SAME direction at 70 km/h and 50 km/h. After how many hours will they be 60 km apart?

- (A) 3
- (B) 2
- (C) 4
- (D) 1.5

Answer: (A)

Solution: Relative speed (same direction) = $70 - 50 = 20$ km/h. Time = $60/20 = 3$ hours.

[New Practice]

Q17. [NAT · 2 marks]

A boat travels downstream at 18 km/h and upstream at 10 km/h. Find the speed of the boat in still water.

Answer: 14

Solution: $(18+10)/2 = 14$ km/h.

[New Practice]

Q18. [NAT · 1 mark]

Using the previous question's data, find the speed of the current.

Answer: 4

Solution: $(18-10)/2 = 4$ km/h.

[New Practice]

Q19. [MSQ · 2 marks]

Which of the following are TRUE about average speed calculations?

- (A) For a journey covering equal DISTANCES at two different speeds, the average speed is the HARMONIC mean of the two speeds, not the arithmetic mean
- (B) Average speed = Total distance / Total time, always
- (C) If speed is constant throughout, the average speed equals that constant speed
- (D) Average speed can always be computed as the simple arithmetic mean of the speeds involved, regardless of the distances/times at each speed

Answer: (A), (B), (C)

Solution: (D) is false in general -- the simple arithmetic mean of speeds is only valid when each speed is maintained for an EQUAL TIME, not equal distance.

[New Practice]

Q20. [NAT · 2 marks]

A train 150 metres long crosses a platform 250 metres long in 20 seconds. Find the speed of the train in m/s.

Answer: 20

Solution: Total distance covered = $150+250 = 400$ m. Speed = $400/20 = 20$ m/s.

Topic 5: Logarithmic Loops Complexity

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [MCQ · 1 mark]

A loop of the form `for(i=1; i<=n; i=i*2)` runs approximately how many times, as a function of n ?

- (A) n
- (B) $\log_2 n$
- (C) $n/2$
- (D) n^2

Answer: (B)

[GATE-style]

Q2. [NAT · 2 marks]

For `i=1; while(i<=n){ i=i*2; }` with $n=64$, how many iterations does the loop body execute?

Answer: 7

Solution: i takes values 1, 2, 4, 8, 16, 32, 64 (each satisfying $i \leq 64$, so the body runs), producing $i=128$ which fails the condition. Total iterations = $7 = \log_2(64) + 1 = 6 + 1$.

[GATE-style]

Q3. [MCQ · 1 mark]

A loop `for(i=n; i>=1; i=i/2)` (integer division) runs approximately how many times?

- (A) n
- (B) $\log_2 n$
- (C) $n/2$
- (D) $2n$

Answer: (B)

[GATE-style]

Q4. [MCQ · 1 mark]

The time complexity of a single loop that halves (or doubles) its counter each iteration is:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n \log n)$
- (D) $O(n^2)$

Answer: (B)

[GATE-style]

Q5. [NAT · 2 marks]

For `for(i=1; i<n; i=i*3)` with $n=100$, approximately how many iterations occur? (Give $\lceil \log_3(100) \rceil$.)

Answer: 5

Solution: $\log_3(100) = \ln(100)/\ln(3) \approx 4.605/1.0986 \approx 4.19$, so $\lceil 4.19 \rceil = 5$.

[GATE-style]

Q6. [MCQ · 1 mark]

Binary search's core "halving" loop/recursion on an array of size n has time complexity:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n \log n)$
- (D) $O(\sqrt{n})$

Answer: (B)

[GATE-style]

Q7. [MSQ · 2 marks]

Which of the following loop patterns have $O(\log n)$ time complexity?

- (A) `for(i=1; i<=n; i=i*2)`
- (B) `for(i=n; i>=1; i=i/2)`
- (C) `for(i=1; i<=n; i++)`
- (D) `for(i=1; i<=n; i=i+2)`

Answer: (A), (B)

Solution: (C) and (D) both increment i by a constant amount each time (1 or 2 respectively), giving $O(n)$, not $O(\log n)$.

[GATE-style]

Q8. [NAT · 1 mark]

For a loop `for(i=1; i<=n; i=i*2)` with $O(1)$ work per iteration, if $n=1024$ (2^{10}), how many total iterations occur ($\log_2(1024)+1$)?

Answer: 11

Solution: $\log_2(1024) = 10$, so total iterations = $10+1 = 11$.

[GATE-style]

Q9. [MCQ · 2 marks]

A nested structure `for(i=1; i<=n; i=i*2) for(j=1; j<=i; j++)` has overall time complexity:

- (A) $O(\log n)$
- (B) $O(n)$
- (C) $O(n \log n)$
- (D) $O(n^2)$

Answer: (B)

Solution: The outer loop runs $O(\log n)$ times with i taking values $1, 2, 4, \dots, n$. For a given i , the inner loop runs i times. Total = $1+2+4+\dots+n \approx 2n-1 = O(n)$ (a geometric series dominated by its largest term, n).

[GATE-style]

Q10. [MCQ · 1 mark]

Repeated squaring (computing x^n via $x, x^2, x^4, x^8, \dots$ by repeated squaring) uses only $O(\log n)$ multiplications instead of the naive $O(n)$ because:

- (A) Each squaring step doubles the effective exponent covered, so only $\log_2 n$ squaring steps are needed to reach exponent n
- (B) It avoids all multiplication entirely
- (C) It uses a lookup table
- (D) It is not actually faster than the naive method

Answer: (A)

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 2 marks]

For `i=1; while(i<n){ i=i*2; }` with `n=50`, how many iterations execute?

Answer: 6

Solution: `i` takes values 1,2,4,8,16,32 (each < 50 , so the body runs), producing `i=64` which fails the condition ($64 < 50$ is false). Total = 6 iterations.

[New Practice]

Q12. [MCQ · 1 mark]

A loop `for(i=1; i<=n; i=i*4)` runs approximately how many times, as a function of `n`?

- (A) $n/4$
- (B) $\log_4 n$
- (C) $4n$
- (D) n

Answer: (B)

[New Practice]

Q13. [NAT · 2 marks]

For `for(i=1; i<=n; i=i*4)` with `n=64`, how many iterations occur?

Answer: 4

Solution: `i` takes values 1,4,16,64 (each ≤ 64 , so the body runs), producing `i=256` which fails. Total = 4
 $= \log_4(64) + 1 = 3 + 1$.

[New Practice]

Q14. [MCQ · 1 mark]

The time complexity of finding an element via repeated halving of a sorted search space (as in binary search) is asymptotically the SAME as the time complexity of:

- (A) A single loop doubling its counter each iteration until it exceeds `n`
- (B) A single loop incrementing its counter by 1 each iteration
- (C) Two nested loops each running `n` times
- (D) A loop running n^2 times

Answer: (A)

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following are TRUE about $O(\log n)$ algorithms/loops?

- (A) They scale very well as `n` grows extremely large, since the number of operations grows very slowly
- (B) Binary search and repeated halving/doubling loops are classic examples
- (C) An $O(\log n)$ algorithm will always run in less real-world time than any $O(n)$ algorithm, for every possible input size

(D) The base of the logarithm (2, 10, e, etc.) does not affect the asymptotic complexity class, only a constant factor

Answer: (A), (B), (D)

Solution: (C) is false -- for very small n , constant-factor overheads can make an $O(\log n)$ algorithm slower in absolute time than a simple $O(n)$ one; the advantage is only guaranteed asymptotically for sufficiently large n .

[New Practice]

Q16. [NAT · 2 marks]

For `for(i=n; i>1; i=i/3)` (integer division) with $n=81$, how many iterations occur?

Answer: 4

Solution: i takes values 81, 27, 9, 3 (each >1 , so the body runs), producing $i=1$ which fails ($1>1$ is false).
Total = 4 = $\log_3(81)$.

[New Practice]

Q17. [MCQ · 1 mark]

Which best explains why `for(i=1; i<=n; i=i+1)` is $O(n)$ while `for(i=1; i<=n; i=i*2)` is $O(\log n)$?

- (A) The first loop's counter grows arithmetically (by a constant amount), while the second grows geometrically (by a constant factor), reaching n exponentially faster
- (B) There is no real difference between the two
- (C) The second loop performs more work per iteration
- (D) The first loop is always faster in practice

Answer: (A)

[New Practice]

Q18. [NAT · 2 marks]

A loop repeatedly halves n until it reaches 1, starting from $n=1000$. Approximately how many halving steps are needed? (Give $\lceil \log_2(1000) \rceil$.)

Answer: 10

Solution: $\log_2(1000) \approx 9.97$, so $\lceil 9.97 \rceil = 10$.

[New Practice]

Q19. [MCQ · 1 mark]

Which of the following is an example of an algorithm with $O(\log n)$ time complexity (excluding any linear pre-processing)?

- (A) Linear search in an unsorted array
- (B) Binary search in a sorted array
- (C) Bubble sort
- (D) Computing the sum of all elements in an array

Answer: (B)

[New Practice]

Q20. [MCQ · 2 marks]

A loop `for(i=1; i<=n; i=i*2)` containing an INDEPENDENT inner loop `for(j=1; j<=n; j++)` (not depending on i) has overall time complexity:

- (A) $O(\log n)$

- (B) $O(n)$
- (C) $O(n \log n)$
- (D) $O(n^2)$

Answer: (C)

Solution: The outer loop runs $O(\log n)$ times, and for EACH iteration the independent inner loop runs $O(n)$ times (unlike the case where the inner bound depends on i) $\Rightarrow O(n \log n)$ total.

Topic 6: Binary Search - 3 Problems + Complexity Variants

This topic includes three fully worked binary-search TRACING problems (finding the number of comparisons for specific targets in a given sorted array), spread across the sections below, plus several classic complexity-variant questions (successful search, unsuccessful search, recursive vs iterative space, and the Master-theorem recurrence).

Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE CSE 1994, classic]

Q1. [MCQ · 1 mark]

The recurrence relation that arises in relation to the complexity of binary search is:

- (A) $T(n) = T(n/2) + k$, where k is a constant
- (B) $T(n) = 2T(n/2) + k$
- (C) $T(n) = T(n-1) + k$
- (D) $T(n) = T(n/2) + n$

Answer: (A)

Solution: Binary search discards half the search space using $O(1)$ work (one comparison) per step, giving the recurrence $T(n) = T(n/2) + k$, which solves to $T(n) = O(\log n)$.

[VERIFIED PYQ — GATE CSE 2007, classic]

Q2. [MCQ · 2 marks]

Consider inserting an element into an array-represented Max-Heap. If a BINARY SEARCH is performed along the path from the new leaf to the root (a path of length $O(\log n)$) to locate the correct insertion position, the number of COMPARISONS performed is:

- (A) $\Theta(\log_2 n)$
- (B) $\Theta(\log_2 \log_2 n)$
- (C) $\Theta(n)$
- (D) $\Theta(n \log_2 n)$

Answer: (B)

Solution: The leaf-to-root path has length $O(\log n)$. Performing binary search along this path of length $O(\log n)$ requires only $O(\log(\log n))$ comparisons (though the data movement/swaps along the path still take $O(\log n)$ time).

[GATE-style, Binary Search Problem 1]

Q3. [NAT · 1 mark]

Given the sorted array $[2, 5, 8, 12, 16, 23, 38, 45, 56, 72, 91]$ (0-indexed, size 11), find the number of comparisons made by standard binary search (comparing against the middle element each time) to find the target value 23.

Answer: 1

Solution: $\text{low}=0$, $\text{high}=10$, $\text{mid}=5$, $\text{arr}[5]=23 = \text{target}$. Found on the very first comparison.

[GATE-style, Binary Search Problem 1 continued]

Q4. [NAT · 2 marks]

Using the same array $[2, 5, 8, 12, 16, 23, 38, 45, 56, 72, 91]$, find the number of comparisons to search for the value 91 (the last element).

Answer: 4

Solution: low=0,high=10,mid=5,arr[5]=23<91⇒low=6 [1].

low=6,high=10,mid=8,arr[8]=56<91⇒low=9 [2]. low=9,high=10,mid=9,arr[9]=72<91⇒low=10 [3].

low=10,high=10,mid=10,arr[10]=91=target [4]. Total = 4 comparisons.

[GATE-style, Binary Search Problem 2]

Q5. [MCQ · 1 mark]

For a sorted array of $n=15$ elements, what is the MAXIMUM number of comparisons required by binary search in the worst case (using $\lfloor \log_2 n \rfloor + 1$)?

- (A) 4
- (B) 3
- (C) 5
- (D) 15

Answer: (A)

Solution: $\lfloor \log_2(15) \rfloor + 1 = 3 + 1 = 4$.

[GATE-style, Binary Search Problem 2 continued]

Q6. [NAT · 2 marks]

For a sorted array of $n=1000$ elements, find the maximum number of comparisons required by binary search in the worst case, using $\lfloor \log_2 n \rfloor + 1$.

Answer: 10

Solution: $\log_2(1000) \approx 9.966$, floor = 9, +1 = 10.

[GATE-style, Binary Search Problem 3]

Q7. [MCQ · 2 marks]

Binary search is performed on a sorted array of n elements to search for a key that is NOT present, falling between positions m and $m+1$. In the worst case, how many comparisons are needed to determine that the key is absent?

- (A) $\lfloor \log_2(n+1) \rfloor$
- (B) n
- (C) $\log_2 n - 1$
- (D) $n/2$

Answer: (A)

Solution: This is the classic result for unsuccessful search -- the number of comparisons equals the height of the implicit binary-search decision tree, $\lfloor \log_2(n+1) \rfloor$.

[GATE-style, Binary Search Problem 3 continued]

Q8. [NAT · 1 mark]

For an unsuccessful binary search on a sorted array of $n=15$ elements, find the worst-case number of comparisons using $\lfloor \log_2(n+1) \rfloor$.

Answer: 4

Solution: $\log_2(16) = 4$ exactly.

[GATE-style]

Q9. [MCQ · 1 mark]

The space complexity of the standard ITERATIVE implementation of binary search is:

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n \log n)$

Answer: (A)

[GATE-style]

Q10. [MCQ · 1 mark]

The space complexity of the standard RECURSIVE implementation of binary search (due to the call stack) is:

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n^2)$

Answer: (B)

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

Given the sorted array [1,3,4,7,9,11,14,18,22] (0-indexed, size 9), find the number of comparisons made by binary search to find the target value 9.

Answer: 1

Solution: low=0,high=8,mid=4,arr[4]=9=target. Found on the first comparison.

[New Practice]

Q12. [NAT · 2 marks]

Using the same array [1,3,4,7,9,11,14,18,22], find the number of comparisons to search for the value 22.

Answer: 4

Solution: low=0,high=8,mid=4,arr[4]=9<22⇒low=5 [1]. low=5,high=8,mid=6,arr[6]=14<22⇒low=7 [2]. low=7,high=8,mid=7,arr[7]=18<22⇒low=8 [3]. low=8,high=8,mid=8,arr[8]=22=target [4]. Total = 4.

[New Practice]

Q13. [MCQ · 1 mark]

For a sorted array of $n=100$ elements, find the maximum number of comparisons in the worst case using $\lfloor \log_2 n \rfloor + 1$.

- (A) 7
- (B) 6
- (C) 8
- (D) 100

Answer: (A)

Solution: $\lfloor \log_2(100) \rfloor + 1 = 6 + 1 = 7$.

[New Practice]

Q14. [NAT · 1 mark]

For a sorted array of $n=500$ elements, find the maximum worst-case comparisons using $\lfloor \log_2 n \rfloor + 1$.

Answer: 9

Solution: $\log_2(500) \approx 8.966$, floor = 8, +1 = 9.

[New Practice]

Q15. [MCQ · 2 marks]

A search key falls between two elements of a sorted, key-absent array of $n=31$ elements. Using $\lfloor \log_2(n+1) \rfloor$, find the worst-case number of comparisons.

- (A) 5
- (B) 4
- (C) 6
- (D) 31

Answer: (A)

Solution: $\log_2(32) = 5$ exactly.

[New Practice]

Q16. [MSQ · 2 marks]

Which of the following are TRUE about binary search?

- (A) It requires the array to be sorted beforehand
- (B) Its worst-case time complexity is $O(\log n)$
- (C) It can be implemented both iteratively and recursively
- (D) It always performs better than linear search, even on small unsorted arrays

Answer: (A), (B), (C)

Solution: (D) is false -- binary search REQUIRES a sorted array; it does not work correctly on an unsorted one at all, so the comparison in that scenario is not meaningful.

[New Practice]

Q17. [MCQ · 2 marks]

In the recurrence $T(n) = T(n/2) + k$ for binary search, solving via the Master theorem ($a=1$, $b=2$) gives:

- (A) $T(n) = \Theta(n)$
- (B) $T(n) = \Theta(\log n)$
- (C) $T(n) = \Theta(n \log n)$
- (D) $T(n) = \Theta(n^2)$

Answer: (B)

Solution: $a=1$, $b=2$, $n^{\log_b a} = n^0 = 1$. $f(n) = k = \Theta(n^0) \Rightarrow$ Master theorem Case 2 $\Rightarrow T(n) = \Theta(\log n)$.

[New Practice]

Q18. [NAT · 1 mark]

Given a sorted array of size $n=63$, find the maximum number of comparisons in the worst case for a successful search, using $\lfloor \log_2 n \rfloor + 1$.

Answer: 6

Solution: $\log_2(63) \approx 5.977$, floor = 5, +1 = 6.

[New Practice]

Q19. [MCQ · 2 marks]

Ternary search (dividing the search space into 3 parts instead of 2 at each step) has time complexity:

- (A) $O(\log_3 n)$, which remains $O(\log n)$ asymptotically, just with a different (larger) constant factor than binary search's $\log_2 n$
- (B) $O(\log_2 n)$, identical constant factor to binary search
- (C) $O(n)$
- (D) $O(n \log n)$

Answer: (A)

Solution: Ternary search performs more comparisons per level (2 instead of 1) but needs fewer levels ($\log_3 n$ instead of $\log_2 n$); it remains $\Theta(\log n)$ overall but is NOT actually faster than binary search in practice, due to the extra comparisons per level.

[New Practice]

Q20. [NAT · 2 marks]

For a sorted array of $n=2,000,000$ (2 million) elements, find the maximum number of comparisons using $\lfloor \log_2 n \rfloor + 1$.

Answer: 21

Solution: $\log_2(2000000) \approx 20.93$, floor = 20, +1 = 21.

Topic 7: Uninformed Search - DLS & IDS

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [MCQ · 1 mark]

Depth-Limited Search (DLS) is essentially:

- (A) BFS with a maximum depth bound
- (B) DFS with a predetermined maximum depth limit ℓ , beyond which nodes are not expanded
- (C) A* search without a heuristic
- (D) UCS with a cost limit instead of a depth limit

Answer: (B)

[GATE-style]

Q2. [MCQ · 1 mark]

In DLS, nodes at the depth limit ℓ are treated as if:

- (A) They are automatically goal nodes
- (B) They have no successors (the search does not expand them further)
- (C) They should be immediately removed from the tree
- (D) They are always non-goal nodes with infinite cost

Answer: (B)

[GATE-style]

Q3. [MCQ · 1 mark]

DLS can fail to find a solution even if one exists, specifically when:

- (A) The depth limit ℓ is set too small, and the actual solution lies at a depth greater than ℓ
- (B) The depth limit ℓ is set larger than necessary
- (C) The branching factor is 1
- (D) The graph has no cycles

Answer: (A)

[GATE-style]

Q4. [MCQ · 1 mark]

The time complexity of Depth-Limited Search with depth limit ℓ and branching factor b is:

- (A) $O(b^\ell)$
- (B) $O(\ell^b)$
- (C) $O(b\ell)$
- (D) $O(\log \ell)$

Answer: (A)

[GATE-style]

Q5. [MCQ · 2 marks]

The space complexity of Depth-Limited Search with depth limit ℓ and branching factor b is:

- (A) $O(b^\ell)$
- (B) $O(b\ell)$

- (C) $O(l)$
- (D) $O(b)$

Answer: (B)

Solution: Like plain DFS, DLS only needs to store the current path plus the unexplored siblings along that path, giving $O(bl)$ space -- far less than the $O(b^l)$ needed to store the entire tree.

[GATE-style]

Q6. [MCQ · 1 mark]

Iterative Deepening Search (IDS / IDDFS) works by:

- (A) Running plain BFS once, then plain DFS once
- (B) Repeatedly running Depth-Limited Search with an increasing depth limit (0,1,2,...) until a solution is found
- (C) Running DLS only once, with the maximum possible depth limit
- (D) Combining A* with UCS

Answer: (B)

[GATE-style]

Q7. [MCQ · 1 mark]

IDS combines which two properties, from BFS and DFS respectively?

- (A) BFS's low memory usage and DFS's completeness
- (B) BFS's completeness/optimalty (for uniform step costs) and DFS's low memory usage
- (C) BFS's high memory usage and DFS's incompleteness
- (D) Neither's properties; it has entirely new properties

Answer: (B)

[GATE-style]

Q8. [NAT · 2 marks]

In IDS, depth-limited search is repeated for depth limits 0, 1, 2, 3. The single root node (at depth 0) is regenerated once at the start of every one of these iterations. How many times, in total, is the root node generated across all 4 iterations?

Answer: 4

Solution: The root is regenerated once per iteration; with 4 iterations (limits 0-3), it is generated 4 times.

[GATE-style]

Q9. [MSQ · 2 marks]

Which of the following are TRUE about Iterative Deepening Search?

- (A) It is complete, provided the branching factor is finite
- (B) It is optimal when all step costs are equal (like BFS)
- (C) It has the same asymptotic time complexity as BFS, $O(b^d)$, despite the seemingly wasteful re-exploration of shallow nodes
- (D) Its space complexity is $O(b^d)$, same as BFS

Answer: (A), (B), (C)

Solution: (D) is false -- IDS's key advantage is LOW memory usage, $O(bd)$, just like plain DFS, NOT $O(b^d)$ like BFS.

[GATE-style]

Q10. [MCQ · 2 marks]

The overhead (extra nodes generated) of IDS compared to a single BFS run to the same depth d is generally considered:

- (A) Prohibitively large, making IDS impractical
- (B) Relatively small (a modest constant-factor overhead) for exponential-branching-factor trees, since the cost is dominated by the last (deepest) iteration
- (C) Exactly zero
- (D) Larger than the cost of BFS itself, making IDS strictly worse

Answer: (B)

Solution: Because tree growth is exponential in depth, the nodes generated by all the shallower "wasted" iterations combined remain small relative to the final (deepest) iteration, so the overhead is modest for reasonably large branching factors.

Section B — New Practice Questions (10)

[New Practice]

Q11. [MCQ · 1 mark]

Which of the following is TRUE about DLS's completeness?

- (A) DLS is always complete, regardless of the depth limit chosen
- (B) DLS is complete only if a solution exists at or before the chosen depth limit ℓ
- (C) DLS is never complete
- (D) DLS's completeness does not depend on the depth limit at all

Answer: (B)

[New Practice]

Q12. [MCQ · 1 mark]

Which of the following is TRUE about DLS's optimality?

- (A) DLS is always optimal
- (B) DLS is not guaranteed to be optimal, even if it finds a solution, since path costs might not correlate with depth
- (C) DLS is optimal only when using a heuristic
- (D) DLS's optimality depends only on the depth limit being very large

Answer: (B)

[New Practice]

Q13. [NAT · 1 mark]

IDS is run with branching factor $b=3$, up to depth limit $d=3$. At the final (deepest) iteration alone, how many nodes are generated at depth 3 (i.e. b^d)?

Answer: 27

Solution: $3^3 = 27$.

[New Practice]

Q14. [MCQ · 1 mark]

Compared to plain (unbounded) DFS, DLS's main advantage is:

- (A) Guaranteed optimality

- (B) A termination guarantee -- DLS is guaranteed to terminate (even without a solution) since it will not explore beyond depth ℓ , unlike plain DFS which may run forever on an infinite/very deep branch
- (C) Lower time complexity
- (D) No need for a visited-set ever

Answer: (B)

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following are TRUE about choosing the depth limit ℓ for DLS?

- (A) If ℓ is set too small, a valid solution deeper than ℓ will be missed
- (B) If ℓ is set too large, DLS behaves increasingly like plain (unbounded) DFS, potentially losing its efficiency/termination advantages
- (C) The ideal ℓ exactly equals the depth of the shallowest goal, but this is often unknown in advance
- (D) The depth limit has no effect on the outcome of the search

Answer: (A), (B), (C)

Solution: (D) is false -- the depth limit is precisely the parameter DLS depends on for both correctness and efficiency.

[New Practice]

Q16. [NAT · 2 marks]

IDS with branching factor $b=2$ runs depth-limited searches for limits 0, 1, 2. Find the total number of nodes generated by the FINAL iteration alone (limit=2), i.e. $b^0 + b^1 + b^2$.

Answer: 7

Solution: $1+2+4 = 7$.

[New Practice]

Q17. [MCQ · 2 marks]

Why is IDS generally preferred over plain BFS when memory is a serious constraint, despite IDS re-exploring shallow nodes multiple times?

- (A) IDS uses far less memory ($O(bd)$ vs $O(b^d)$ for BFS) while retaining the same completeness and optimality guarantees for uniform step costs
- (B) IDS is always faster in wall-clock time than BFS
- (C) IDS never re-explores any node
- (D) BFS cannot guarantee completeness at all

Answer: (A)

[New Practice]

Q18. [MCQ · 1 mark]

If the depth of the shallowest goal, d , is unknown in advance, which uninformed strategy is most naturally suited to still find a solution efficiently in terms of memory?

- (A) Plain BFS, since it always explores the shallowest depth first anyway
- (B) Iterative Deepening Search (IDS), since it progressively increases the depth limit and finds the shallowest goal like BFS, but with DFS-level memory usage
- (C) Plain DFS with an arbitrarily large fixed depth limit
- (D) UCS, since it ignores depth entirely

Answer: (B)

[New Practice]

Q19. [NAT · 2 marks]

IDS with branching factor $b=4$ runs depth-limited searches for limits 0 through 2 (three iterations). Find the number of nodes generated in the LAST (deepest, limit=2) iteration alone, i.e. $b^0 + b^1 + b^2$.

Answer: 21

Solution: $1 + 4 + 16 = 21$.

[New Practice]

Q20. [MCQ · 1 mark]

Which uninformed search strategy guarantees finding a solution at the SHALLOWEST possible depth, while using memory proportional to the DEPTH of that solution (rather than the full breadth of the tree)?

- (A) Plain BFS (uses $O(b^d)$ memory, not $O(d)$)
- (B) Plain DFS (does not guarantee shallowest-depth solution)
- (C) Depth-Limited Search alone, with a correctly guessed limit (still doesn't guarantee shallowest unless the limit exactly equals d)
- (D) Iterative Deepening Search (IDS)

Answer: (D)
